

Track 2F-Bis Harmonic-Offset Probe Campaign Results

Track 2 Campaign Closeout

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the approved `Track 2F-bis` harmonic-offset probe campaign. The campaign goal was to keep the clean `Track 2F` non-harmonic residual-offset baseline alive, then test a harmonic-forced residual-offset branch on the same three required deployment surfaces: `global`, `Fw`, and `Bw`.

The first execution completed the three clean baseline runs and failed the three harmonic-offset runs before training started because the campaign runner did not yet route the new `harmonic_residual_offset_probe` model type. The runner registration has now been fixed, the three failed harmonic configs were repaired through a dedicated rerun, and the effective closeout state is six trained candidates with zero unresolved failed runs.

The scalar leaderboard ranks `Fw` first by `test_mae`, but that ranking remains diagnostic only. Track 2F-bis keeps three parallel branch outcomes for future curve-first verification and deployment analysis: one `global`, one `Fw`, and one `Bw`. The clean and harmonic branches are also both retained because the clean branch is the no-harmonic baseline for later multi-head, composite-loss, and new-index work.

Campaign Artifacts

Artifact	Path
Original campaign output	<code>output/training_campaigns/2026-06-04-23-31-57_track2f_bis_harmonic_offset_probe_campaign_2026_06_04</code>
Repair campaign output	<code>output/training_campaigns/2026-06-05-16-07-17_track2f_bis_harmonic_offset_probe_repair_2026_06_05</code>
Original leaderboard	<code>output/training_campaigns/2026-06-04-23-31-57_track2f_bis_harmonic_offset_probe_campaign_2026_06_04/campaign_leaderboard.yaml</code>
Repair leaderboard	<code>output/training_campaigns/2026-06-05-16-07-17_track2f_bis_harmonic_offset_probe_repair_2026_06_05/campaign_leaderboard.yaml</code>
Original execution report	<code>output/training_campaigns/2026-06-04-23-31-57_track2f_bis_harmonic_offset_probe_campaign_2026_06_04/campaign_execution_report.md</code>
Repair execution report	<code>output/training_campaigns/2026-06-05-16-07-17_track2f_bis_harmonic_offset_probe_repair_2026_06_05/campaign_execution_report.md</code>
Planning report	<code>doc/reports/campaign_plans/track2/2026-06-04-22-57-04_track2f_bis_harmonic_offset_probe_campaign_plan_report.md</code>
Campaign technical document	<code>doc/technical/2026-06/2026-06-04/2026-06-04-21-14-52_track2f_bis_harmonic_offset_probe.md</code>
Runner-fix technical document	<code>doc/technical/2026-06/2026-06-05/2026-06-05-15-56-59_track2f_bis_campaign_runner_model_type_fix.md</code>

Execution Summary

Field	Value
Original campaign name	track2f_bis_harmonic_offset_probe_campaign_2026_06_04
Repair campaign name	track2f_bis_harmonic_offset_probe_repair_2026_06_05
Original completed runs	3
Original failed runs	3
Repair completed runs	3
Repair failed runs	0
Effective completed candidates	6
Unresolved failed candidates	0
Clean model type	sequential_residual_offset_probe
Harmonic model type	harmonic_residual_offset_probe
Tested surfaces	global , Fw , Bw
Runner-level scalar first entry	te_track2f_bis_harmonic_residual_offset_fw

Directional Branch Results

Surface	Role	Run	Family	Test MAE	Test RMSE	Val MAE	Params
global	clean baseline	te_track2f_bis_clean_residual_offset_global	track2f_bis_clean_sequential_residual_offset_global	0.003528	0.004010	0.003717	92,802
global	harmonic offset	te_track2f_bis_harmonic_residual_offset_global	track2f_bis_harmonic_residual_offset_global	0.003538	0.003932	0.003659	85,747
Fw	clean baseline	te_track2f_bis_clean_residual_offset_fw	track2f_bis_clean_sequential_residual_offset_fw	0.003446	0.003972	0.003474	92,802
Fw	harmonic offset	te_track2f_bis_harmonic_residual_offset_fw	track2f_bis_harmonic_residual_offset_fw	0.002862	0.003334	0.002941	85,747
Bw	clean baseline	te_track2f_bis_clean_residual_offset_bw	track2f_bis_clean_sequential_residual_offset_bw	0.003540	0.004203	0.003820	92,802
Bw	harmonic offset	te_track2f_bis_harmonic_residual_offset_bw	track2f_bis_harmonic_residual_offset_bw	0.003336	0.003935	0.003555	85,747

These six rows are the closeout result. The harmonic `Fw` row is the runner-level scalar first entry, but it does not replace the required `global` or `Bw` branch candidates.

Clean Versus Harmonic Delta

Surface	Clean MAE	Harmonic MAE	MAE Gain [%]	Clean RMSE	Harmonic RMSE	RMSE Gain
global	0.003528	0.003538	-0.28	0.004010	0.003932	1.94
Fw	0.003446	0.002862	16.96	0.003972	0.003334	16.91
Bw	0.003540	0.003336	5.75	0.004203	0.003935	6.38

The scalar result is direction-dependent. Harmonic forcing improves `Fw` clearly, improves `Bw` moderately, and leaves `global` almost unchanged on `test_mae` while still reducing `test_rmse`. This is enough to keep the harmonic-offset branch alive, but it is not enough to make a deployment decision before the official curve-first Track 2 refresh.

Failure Repair

The first campaign failure was not a model-convergence result. The three harmonic runs stopped immediately because `scripts/training/run_training_campaign.py` did not include `harmonic_residual_offset_probe` in the campaign model-type dispatch map.

The fix adds `harmonic_residual_offset_probe` to the campaign runner and routes it through the existing feedforward training entry point. A one-batch setup validation was then run on the failed global queue config, followed by a repair campaign over the three failed harmonic configs. The repair campaign completed all three runs and produced the expected leaderboard, best-run, family registry, and program-registry artifacts.

The launcher wrapper was also hardened for a recurring operator issue. When the PowerShell launcher is already running inside the requested Conda environment, it now calls `python` directly instead of nesting another `conda run`. This prevents the misleading terminal-level `conda run ... failed` message seen in the previous Track 2F and Track 2F-bis operator runs.

Scalar Leaderboard

Rank	Surface	Run	Family	Test MAE	Test RMSE	Val MAE	Params
1	Fw	<code>te_track2f_bis_harmonic_residual_offset_fw</code>	<code>track2f_bis_harmonic_residual_offset_fw</code>	0.002862	0.003334	0.002941	85,747
2	Bw	<code>te_track2f_bis_harmonic_residual_offset_bw</code>	<code>track2f_bis_harmonic_residual_offset_bw</code>	0.003336	0.003935	0.003555	85,747
3	Fw	<code>te_track2f_bis_clean_residual_offset_fw</code>	<code>track2f_bis_clean_sequential_residual_offset_fw</code>	0.003446	0.003972	0.003474	92,802
4	global	<code>te_track2f_bis_clean_residual_offset_global</code>	<code>track2f_bis_clean_sequential_residual_offset_global</code>	0.003528	0.004010	0.003717	92,802
5	global	<code>te_track2f_bis_harmonic_residual_offset_global</code>	<code>track2f_bis_harmonic_residual_offset_global</code>	0.003538	0.003932	0.003659	85,747
6	Bw	<code>te_track2f_bis_clean_residual_offset_bw</code>	<code>track2f_bis_clean_sequential_residual_offset_bw</code>	0.003540	0.004203	0.003820	92,802

Registry Effects

Registry Scope	Current Family Best	Test MAE	Interpretation
<code>track2f_bis_clean_sequential_residual_offset_global</code>	<code>te_track2f_bis_clean_residual_offset_global</code>	0.003528	Clean global baseline retained
<code>track2f_bis_clean_sequential_residual_offset_fw</code>	<code>te_track2f_bis_clean_residual_offset_fw</code>	0.003446	Clean forward baseline retained
<code>track2f_bis_clean_sequential_residual_offset_bw</code>	<code>te_track2f_bis_clean_residual_offset_bw</code>	0.003540	Clean backward baseline retained
<code>track2f_bis_harmonic_residual_offset_global</code>	<code>te_track2f_bis_harmonic_residual_offset_global</code>	0.003538	Harmonic global branch trained
<code>track2f_bis_harmonic_residual_offset_fw</code>	<code>te_track2f_bis_harmonic_residual_offset_fw</code>	0.002862	Harmonic forward branch trained
<code>track2f_bis_harmonic_residual_offset_bw</code>	<code>te_track2f_bis_harmonic_residual_offset_bw</code>	0.003336	Harmonic backward branch trained

The campaign runner refreshed the family registries for all six branch families and updated the program registry. The program-level scalar best did not move because the existing Wave 2B periodic `GRU` backward-only model is still stronger on scalar `test_mae`.

Track 2 Boundary

Official Track 2 curve-first verification was not run as part of this closeout. Under campaign governance, that remains a separate operator-approved workflow after the final campaign-results report and PDF are complete.

The next verification package must evaluate the clean and harmonic Track 2F-bis branches on `global` , `Fw` , and `Bw` in parallel. The intended decision is not "one best model overall" at this stage. The intended decision is whether the harmonic-offset intervention improves curve following for each deployment surface while preserving the clean non-harmonic baseline as a control.

Closeout Decision

Track 2F-bis is complete from an execution standpoint after repair: all six planned candidates now have successful training artifacts, no unresolved failed run remains, and the active campaign state can be cleared.

From a modeling standpoint, Track 2F-bis is a useful intervention branch but not yet a deployment promotion. The scalar metrics support carrying the harmonic `Fw` and `Bw` branches forward, while the `global` branch needs curve-first verification before any stronger conclusion.

Recommended Follow-Up

1. Keep six Track 2F-bis candidates available: clean and harmonic variants for `global` , `Fw` , and `Bw` .
2. Do not collapse the branch set to the scalar-first harmonic `Fw` candidate.
3. Run the separate official Track 2 refresh once the closeout PDF is accepted.
4. Inspect the curve overlays before deciding whether harmonic-offset should become the next training family or remain a comparison branch.